MSG No. 127.388 $P4/mbm1'$ [Type II, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `4/m..1'`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,6,9,10,11,14,17,18,19,22,25,26,27,30]`
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[4,5,7,8,12,13,15,16,20,21,23,24,28,29,31,32]`

Table 2: Wyckoff site: `2b`, site symmetry: `4/m..1'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,3,6,9,10,11,14,17,18,19,22,25,26,27,30]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[4,5,7,8,12,13,15,16,20,21,23,24,28,29,31,32]`

Table 3: Wyckoff site: `2c`, site symmetry: `m.mm1'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,6,7,8,9,14,15,16,17,22,23,24,25,30,31,32]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[2,3,4,5,10,11,12,13,18,19,20,21,26,27,28,29]`

Table 4: Wyckoff site: `2d`, site symmetry: `m.mm1'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,6,7,8,9,14,15,16,17,22,23,24,25,30,31,32]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[2,3,4,5,10,11,12,13,18,19,20,21,26,27,28,29]`

Table 5: Wyckoff site: `4e`, site symmetry: `4..1'`

No.	position	mapping
1	$[0, 0, z]$	`[1,2,3,6,17,18,19,22]`
2	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[4,5,7,8,20,21,23,24]`
3	$[0, 0, -z]$	`[9,10,11,14,25,26,27,30]`
4	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[12,13,15,16,28,29,31,32]`

Table 6: Wyckoff site: `4f`, site symmetry: `2.mm1'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, z\right]$	`[1,6,15,16,17,22,31,32]`
2	$\left[\frac{1}{2}, 0, z\right]$	`[2,3,12,13,18,19,28,29]`
3	$\left[\frac{1}{2}, 0, -z\right]$	`[4,5,10,11,20,21,26,27]`
4	$\left[0, \frac{1}{2}, -z\right]$	`[7,8,9,14,23,24,25,30]`

Table 7: Wyckoff site: `4g`, site symmetry: `m.2m1'`

No.	position	mapping
1	$\left[x,\, x+\frac{1}{2},\, 0\right]$	`[1,7,14,16,17,23,30,32]`
2	$\left[\frac{1}{2}-x,\, x,\, 0\right]$	`[2,5,11,12,18,21,27,28]`
3	$\left[x+\frac{1}{2},\, -x,\, 0\right]$	`[3,4,10,13,19,20,26,29]`
4	$\left[-x,\, \frac{1}{2}-x,\, 0\right]$	`[6,8,9,15,22,24,25,31]`

Table 8: Wyckoff site: `4h`, site symmetry: `m.2m1'`

No.	position	mapping
1	$\left[x,\, x+\frac{1}{2},\, \frac{1}{2}\right]$	`[1,7,14,16,17,23,30,32]`
2	$\left[\frac{1}{2}-x,\, x,\, \frac{1}{2}\right]$	`[2,5,11,12,18,21,27,28]`
3	$\left[x+\frac{1}{2},\, -x,\, \frac{1}{2}\right]$	`[3,4,10,13,19,20,26,29]`
4	$\left[-x,\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	`[6,8,9,15,22,24,25,31]`

Table 9: Wyckoff site: `8i`, site symmetry: `m..1'`

No.	position	mapping
1	$[x,\, y,\, 0]$	`[1,14,17,30]`
2	$[-y,\, x,\, 0]$	`[2,11,18,27]`
3	$[y,\, -x,\, 0]$	`[3,10,19,26]`
4	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, 0\right]$	`[4,13,20,29]`
5	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, 0\right]$	`[5,12,21,28]`
6	$[-x,\, -y,\, 0]$	`[6,9,22,25]`
7	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, 0\right]$	`[7,16,23,32]`
8	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, 0\right]$	`[8,15,24,31]`

Table 10: Wyckoff site: `8j`, site symmetry: `m..1'`

No.	position	mapping
1	$\left[x,\, y,\, \frac{1}{2}\right]$	`[1,14,17,30]`
2	$\left[-y,\, x,\, \frac{1}{2}\right]$	`[2,11,18,27]`
3	$\left[y,\, -x,\, \frac{1}{2}\right]$	`[3,10,19,26]`
4	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}\right]$	`[4,13,20,29]`
5	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}\right]$	`[5,12,21,28]`
6	$\left[-x,\, -y,\, \frac{1}{2}\right]$	`[6,9,22,25]`
7	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{2}\right]$	`[7,16,23,32]`
8	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	`[8,15,24,31]`

Table 11: Wyckoff site: `8k`, site symmetry: `..m1'`

No.	position	mapping
1	$\left[x,\, x+\frac{1}{2},\, z\right]$	`[1,16,17,32]`
2	$\left[\frac{1}{2}-x,\, x,\, z\right]$	`[2,12,18,28]`

continued ...

Table 11

No.	position	mapping
3	$\left[x+\frac{1}{2},\,-x,\,z\right]$	[3,13,19,29]
4	$\left[x+\frac{1}{2},\,-x,\,-z\right]$	[4,10,20,26]
5	$\left[\frac{1}{2}-x,\,x,\,-z\right]$	[5,11,21,27]
6	$\left[-x,\,\frac{1}{2}-x,\,z\right]$	[6,15,22,31]
7	$\left[x,\,x+\frac{1}{2},\,-z\right]$	[7,14,23,30]
8	$\left[-x,\,\frac{1}{2}-x,\,-z\right]$	[8,9,24,25]

Table 12: Wyckoff site: 16l, site symmetry: 11'

No.	position	mapping
1	$[x,\,y,\,z]$	[1,17]
2	$[-y,\,x,\,z]$	[2,18]
3	$[y,\,-x,\,z]$	[3,19]
4	$\left[x+\frac{1}{2},\,\frac{1}{2}-y,\,-z\right]$	[4,20]
5	$\left[\frac{1}{2}-x,\,y+\frac{1}{2},\,-z\right]$	[5,21]
6	$[-x,\,-y,\,z]$	[6,22]
7	$\left[y+\frac{1}{2},\,x+\frac{1}{2},\,-z\right]$	[7,23]
8	$\left[\frac{1}{2}-y,\,\frac{1}{2}-x,\,-z\right]$	[8,24]
9	$[-x,\,-y,\,-z]$	[9,25]
10	$[y,\,-x,\,-z]$	[10,26]
11	$[-y,\,x,\,-z]$	[11,27]
12	$\left[\frac{1}{2}-x,\,y+\frac{1}{2},\,z\right]$	[12,28]
13	$\left[x+\frac{1}{2},\,\frac{1}{2}-y,\,z\right]$	[13,29]
14	$[x,\,y,\,-z]$	[14,30]
15	$\left[\frac{1}{2}-y,\,\frac{1}{2}-x,\,z\right]$	[15,31]
16	$\left[y+\frac{1}{2},\,x+\frac{1}{2},\,z\right]$	[16,32]